

gz sim process

Entity Component System (ECS)

every object is an Entity (integer ID)

properties are Components attached to entities

Joint entity has: pose, velocity, force components

Link entity has: inertia, collision, visual components

Sensor entity has: imu data component

↓ each sim step

Physics system

DART engine, steps time

Sensor system

reads components, gz topics



gz_ros2_control plugin (also a Gazebo system)

loaded because URDF has `<plugin filename="gz_ros2_control-system">`

configure():

parses `<ros2_control>` block from `robot_description`

for each declared joint: finds the Gazebo entity by name

stores entity IDs so it can read/write them each tick

read()
joint pos/vel
write()
cmd positions

Each sim step calls the plugin's update()

data from ECS → fill StateInterfaces → run controllers → write CommandInterfaces → apply to ECS

ros2_control

ResourceManager

StateInterface slots

CommandInterface slots

Controllers

leg, arm, head, jsb